

Multi-Head Attention (MHA)

Transformer Notes — Concept, Mathematics, Shapes, Intuition & Interview Preparation

1. What is Multi-Head Attention?

Multi-Head Attention is an extension of scaled dot-product attention in which several attention mechanisms operate in parallel. Each head uses its own learned Query, Key and Value projections. The outputs of all heads are concatenated and passed through a final learned linear projection.

Core intuition: same input → different learned views → attention independently → combine the views → final projection.

Input X
↓
Multiple attention heads Head 1 Head 2 ... Head h
↓
Concatenate head outputs
↓
Linear projection W^O
↓
MHA Output

2. Why do we need multiple heads?

A sentence can contain many relationships at the same time. For example: "The animal did not cross the road because it was tired." Understanding "it" may require relating it to "animal", while other attention patterns may capture syntax, nearby context, or semantic relationships.

- Different heads can learn different token-to-token relationships.
- Different heads can represent information in different learned subspaces.
- The roles of heads are not hard-coded; specialization, when it occurs, is learned during training.

3. Single-Head Attention: the foundation

Before MHA, remember scaled dot-product attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{d_k}) \mathbf{V}$$

Query (Q): what am I looking for?

Key (K): what information do I contain / how relevant am I?

Value (V): what information should I provide?

4. What changes in Multi-Head Attention?

Suppose $d_{\text{model}} = 512$ and there are $h = 8$ heads. A common design uses $d_{\text{head}} = 512 / 8 = 64$. Each head therefore performs attention in a 64-dimensional projected space.

Important: do not think of this merely as permanently cutting the original 512-dimensional embedding into eight fixed pieces. Each head has its own learned projections, so every head can form a different 64-dimensional representation from the same input.

5. Q, K and V projections for each head

For head i :

$$\mathbf{Q}_i = \mathbf{XW}_i^{\mathbf{Q}}$$

$$\mathbf{K}_i = \mathbf{XW}_i^{\mathbf{K}}$$

$$\mathbf{V}_i = \mathbf{XW}_i^{\mathbf{V}}$$

The matrices $\mathbf{W}_i^{\mathbf{Q}}$, $\mathbf{W}_i^{\mathbf{K}}$ and $\mathbf{W}_i^{\mathbf{V}}$ are learned parameters. Different heads have different projection matrices.

Same input X		
↓	↓	↓
Head 1 projections	Head 2 projections	Head h projections
$\mathbf{W}_1^{\mathbf{Q}}, \mathbf{W}_1^{\mathbf{K}}, \mathbf{W}_1^{\mathbf{V}}$	$\mathbf{W}_2^{\mathbf{Q}}, \mathbf{W}_2^{\mathbf{K}}, \mathbf{W}_2^{\mathbf{V}}$	$\mathbf{W}_h^{\mathbf{Q}}, \mathbf{W}_h^{\mathbf{K}}, \mathbf{W}_h^{\mathbf{V}}$
↓	↓	↓
64d representations	64d representations	64d representations

6. Attention inside each head

Each head now performs ordinary scaled dot-product attention:

$$\text{head}_i = \text{softmax}(\mathbf{Q}_i \mathbf{K}_i^{\mathbf{T}} / \sqrt{d_k}) \mathbf{V}_i$$

Thus, Multi-Head Attention is not a new kind of attention formula for each head. It is the same scaled dot-product attention repeated in parallel with different learned projections.

7. Shape walkthrough

Let sequence length be n , $d_{\text{model}} = 512$, number of heads $h = 8$, and $d_{\text{head}} = 64$.

Component	Shape
Input X	$n \times 512$
$\mathbf{Q}_i$	$n \times 64$
$\mathbf{K}_i$	$n \times 64$
$\mathbf{V}_i$	$n \times 64$
$\mathbf{Q}_i \mathbf{K}_i^{\mathbf{T}}$	$n \times n$
Attention weights	$n \times n$
Head output	$n \times 64$
Concatenated heads	$n \times 512$
Final output	$n \times 512$

8. Why is $\mathbf{QK}^{\mathbf{T}}$ an $n \times n$ matrix?

There are n query tokens and n key tokens. Every query compares itself with every key, so the score matrix contains one score for every query-key pair:

$$n \text{ queries} \times n \text{ keys} = n \times n$$

This is also the source of the quadratic dependence on sequence length in standard self-attention: the attention-score calculation has $O(n^2)$ scaling with sequence length.

9. Why divide by $\sqrt{d_k}$?

The dot product QK^T can grow in magnitude as the key/query dimension increases. Very large logits can make softmax extremely peaked, which can lead to poor gradients. Scaling by $\sqrt{d_k}$ keeps the logits in a more manageable range and stabilizes training.

$$\text{Attention}(Q, K, V) = \text{softmax}(QK^T / \sqrt{d_k}) V$$

10. Concatenating the heads

With eight heads, each producing $n \times 64$, concatenation along the embedding dimension gives:

$$64 \times 8 = 512 \rightarrow \text{Concatenated output} = n \times 512$$

Concatenation preserves the information produced by the different heads. The model can then mix these representations using the final projection.

11. Final output projection

After concatenation, apply a learned output matrix W^O :

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) W^O$$

For $d_{\text{model}} = 512$, W^O is typically 512×512 , so the final output remains $n \times 512$.

12. Complete mathematical definition

For every head i :

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

$$\text{Attention}(Q_i, K_i, V_i) = \text{softmax}(Q_i K_i^T / \sqrt{d_k}) V_i$$

$$\text{MHA}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O$$

13. Self-Attention vs Multi-Head Self-Attention

Self-attention describes the source of Q , K and V : all three are derived from the same sequence representation X . Multi-head describes using several independent learned projections and attention operations in parallel.

$$\text{Self-attention: } Q = XW^Q, K = XW^K, V = XW^V$$

$$\text{Multi-head self-attention: head}_i = \text{Attention}(XW_i^Q, XW_i^K, XW_i^V)$$

14. Where MHA appears in the Transformer

Encoder: Multi-Head Self-Attention $\rightarrow$ Add & LayerNorm $\rightarrow$ Feed-Forward Network $\rightarrow$ Add & LayerNorm.

Decoder: Masked Multi-Head Self-Attention $\rightarrow$ Add & LayerNorm $\rightarrow$ Multi-Head Cross-Attention $\rightarrow$ Add & LayerNorm $\rightarrow$ Feed-Forward Network $\rightarrow$ Add & LayerNorm.

In cross-attention, Q comes from the decoder while K and V come from the encoder. This lets the decoder use the encoder's representation while generating output.

15. Masking is separate from multi-head attention

MHA itself does not inherently mean masking. In decoder self-attention, a causal mask is added to prevent a position from attending to future positions. The mask is applied to the attention logits before softmax.

	I	am	going	to
I	✓	✗	✗	✗
am	✓	✓	✗	✗

going	✓	✓	✓	✗
to	✓	✓	✓	✓

16. Computational perspective

For fixed d_{model} , splitting the representation across h heads keeps the attention score computation roughly at the same asymptotic order:

One full attention: $O(n^2 d_{\text{model}})$

h heads: $O(h n^2 d_{\text{head}})$

Since $d_{\text{head}} = d_{\text{model}}/h$, this becomes $O(n^2 d_{\text{model}})$.

The major motivation for MHA is therefore richer representation and the ability to learn multiple attention patterns, not simply multiplying the amount of attention computation.

17. Interview-ready answer

Multi-Head Attention is an extension of scaled dot-product attention where multiple attention operations run in parallel using different learned Query, Key and Value projections. Each head operates in a lower-dimensional representation space and can learn different relationships between tokens. The outputs of all heads are concatenated and passed through a final linear projection to produce the output in the model dimension.

18. Five points to remember

- Multiple heads: h attention mechanisms run in parallel.
- Separate projections: every head has its own W_iQ , W_iK and W_iV .
- Independent attention: each head computes scaled dot-product attention.
- Concatenation: head outputs are joined along the embedding dimension.
- Output projection: the concatenated representation is multiplied by W^O .

19. Final mental model

Same input $\rightarrow$ different learned projections $\rightarrow$ separate attention $\rightarrow$ concatenate $\rightarrow$ final linear projection $\rightarrow$ output.

The central intuition: Multi-Head Attention does not merely split the embedding into fixed chunks. It creates multiple learned views of the same input, performs attention in each view, and combines the contextual information.